Homework 6 (Simplex Method)

February 26, 2026

Problems 1-4 are worth 10 points each. Problem 5 is worth 20 points.

1)

Consider the LP

$$\text{minimize } \vec{c} \cdot \vec{x}$$

over polyhedron P. Prove the following:

a)

A feasible solution is optimal if and only if $\vec{c} \cdot \vec{d} \geq 0$ for all feasible directions $\vec{d}$ at $\vec{x}$.

b)

A feasible solution $\vec{x}$ is the unique optimal solution if and only if $\vec{c} \cdot \vec{d} > 0$ for all non-zero feasible directions at $\vec{x}$.

2)

Let $\vec{x}$ be an element of the polyhedron, $P = \left\{ \vec{x} \in \mathbb{R}^n | A\vec{x} = \vec{b}, \vec{x} \geq 0 \right\}$. Prove that $\vec{d} \in \mathbb{R}^n$ is a feasible direction at $\vec{x}$ if and only if $A\vec{d} = 0$ and $d_i \geq 0$ for every i where $x_i = 0$. (Note, we gave some intuition around this in lecture 11. Now we are proving this formally).

3)

Let $\vec{x}$ be a BFS with associated basis matrix, B. If the reduced cost of every non-basic variable is positive, show $\vec{x}$ is the unique optimal solution.

4)

Let $f : \mathbb{Z} \to \mathbb{Z}$ be defined as

$$f(n) = \begin{cases} n^2 & \text{if } n \text{ is even} \\ 2n + 4 & \text{if } n \text{ is odd} \end{cases}$$

Prove or disprove the following:

a)

$f(n) = O(n)$

b)

$f(n) = O\left(n^2\right)$

5)

Consider the LP

$$\text{minimize } -2x_1 - x_2$$

subject to

$$x_1 - x_2 \leq 2$$

$$x_1 + x_2 \leq 6$$

$$x_1, x_2 \geq 0$$

a)

Convert this LP to standard form.

b)

Given the starting BFS where $(x_1, x_2) = (0, 0)$, solve this LP using simplex method. You should show the steps explicitly, but you are encouraged to use a calculator to help with any matrix calculations.

c)

Solve this LP graphically. Draw the path of the basic feasible solutions visited in each step of the simplex method.